

$$\frac{dy}{dx} = -2y \cos(x)$$

Identity $y' = \lambda(x)y$

$$\lambda(x) = -2 \cos(x)$$

Exact Solution $\int y \, dy = \int -2 \cos(x) \, dx$

$$\ln|y| = -2 \sin(x) + C$$

apply initial condition
 $y(0) = y_0 \rightarrow y(x) = y_0 e^{-2 \sin(x)}$

Bounding
Conditions

$$-1 \leq \sin(x) \leq 1$$

$$-2 \leq -2 \sin(x) \leq 2$$

$$e^{-2} \leq e^{-2 \sin(x)} \leq e^2$$

$$y_0 e^{-2} \leq y(x) \leq y_0 e^2$$

Euler's Method
Update

$$y_{n+1} = y_n + h f(x_n, y_n)$$

$$y_{n+1} = y_n + h(-2y_n \cos(x_n))$$

$$y_{n+1} = (1 - 2h \cos(x_n)) y_n$$

$$G(x_n) = 1 - 2h \cos(x_n)$$

Stability Condition

$$|G(x_n)| \leq 1 \quad \leftarrow \text{required for stability}$$

$$\underbrace{|1 - 2h \cos(x)| \leq 1}_{\text{condition}}$$

$$\lambda(x) < 0$$

$$\text{Case 1: } \cos(x) > 0 \quad 0 \leq h \cos(x) \leq 1 \quad h \leq \frac{1}{\cos(x)} \quad \text{stable if } h \leq 1$$

$$\text{Case 2: } \cos(x) = 0 \quad \text{(neutral)} \quad G = 1$$

$$\text{Case 3: } \cos(x) < 0 \quad G = 1 + 2h|\cos(x)| > 1 \quad |G| > 1$$

unstable for any $h > 0$

Conclusion: $|1 - 2h \cos(x)| \leq 1$, but there is no single positive step size h that guarantees stability for all x .

- where $\cos(x) > 0$, euler is stable if $h \leq 1$

- where $\cos(x) < 0$, euler is unstable for any $h > 0$

$\therefore$ since the solution is bounded at $y_0 e^{-2} \leq y(x) \leq y_0 e^2$, the true solution does not blow up. Therefore, any large growth or instability in the Euler approximation is caused by the numerical method, not the differential equation.

This means that the bounding conditions do affect the choice of step size h , and h must be small enough for accuracy. Therefore, Euler's method is conditionally stable.